

A Packing of 13 Four-Spaces in $\mathbb{R}^{13}$

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Abstract

We give 13 4-dimensional subspaces of $\mathbb{R}^{13}$ whose smallest squared chordal distance is at least 2.999701. This cell of the table of best Grassmannian packings has no published value. The best value implied by known packings is 2.981904, obtained by deleting 1 of the 14 four-spaces of the packing found by this search. The value is checked with exact rational arithmetic from integer coordinates. We do not claim it is optimal.

1 Result

Theorem. *There are 13 subspaces of dimension 4 in $\mathbb{R}^{13}$ whose pairwise squared chordal distances $D(a, b) = 4 - \text{tr}(P_a P_b)$ are all at least*

$$D_{\min} = 2.999701 \dots$$

The exact value of $D_{\min}$, a rational number, is given in Appendix A.

2 Context

N. J. A. Sloane's *Table of Best Grassmannian Packings* (last modified 1997), now maintained by Henry Cohn, lists values for some N in the row $m = 13$, $n = 4$. The cell $N = 13$ is blank in Sloane's table, in Sloane's coordinate files and in Cohn's table as checked on October 8, 2026. The best value implied by known packings is 2.981904, obtained by deleting 1 of the 14 four-spaces of the packing found by this search. The Rankin simplex bound for this cell is $4(13 - 4)/13 \cdot 13/12 = 3.000000$, and the value above lies below it.

3 Verification

Each subspace is given by an integer $m \times n$ basis Y (coordinates times 10^{12} , rounded). With $G = Y^\top Y$ the projection is $P = YG^{-1}Y^\top$, so for two subspaces

$$\text{tr}(P_a P_b) = \frac{\text{tr}(\text{adj}(G_a) A \text{adj}(G_b) A^\top)}{\det G_a \det G_b}, \quad A = Y_a^\top Y_b,$$

an exact rational number. No orthonormality is assumed and no floating point is used. Run

`python grassmann/verify.py grassmann/packings/m13n4N13.txt`

in <https://github.com/Ge-limin/math>.

*Found and checked by Claude, a model released by Anthropic, running on the author's laptop. The author sent the prompts.

4 How it was found

Many random starts of gradient descent on the Grassmannian against a log-sum-exp of the pairwise overlaps, sharpened over time, then a polishing stage (`grassmann/solve.py`); about 63 seconds on one laptop, with about ten such searches running at once.

A Exact value

numerator:

```
374962708290493769731895527147685539481046105766247179357099587535368672
230895301602432224716973134255229332139953594556550097270880731023780623
193338144948595294421392174334004885398918272821
```

denominator:

```
124999999999715975133460332685457612379318527820848006402356429179867319
359599615640011681510402025643919983028827701883119070556126664777143386
873375152541245594708579995031280514687656279872
```

B Coordinates

13 blocks of 13 rows with 4 integers each; divide by 10^{12} . File: `grassmann/packings/m13n4N13.txt`.

```
-3520835600 12554861378 487172515118 269481749219 119719840270 -62818165643 10275865719 82468676628 323858626235 -161958831075 189155309912 268731320174
309841049689 61386769982 249171595689 -264280364408 99254893084 151164106332 339156664756 666750643932 208127187941 322667548020 171538568109 -392815006592
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